

Machine Vision

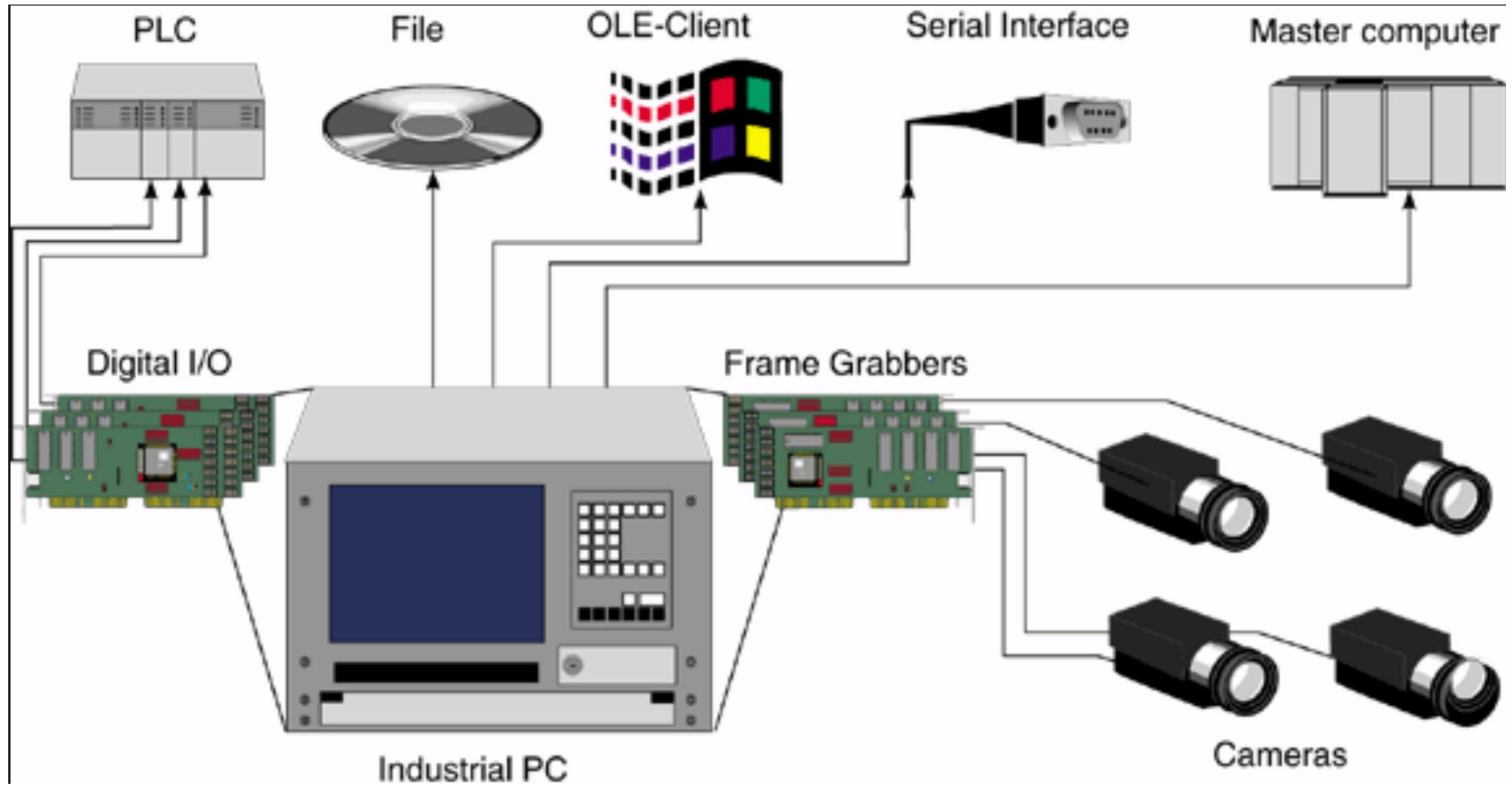
- Introduction -

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Machine Vision Components

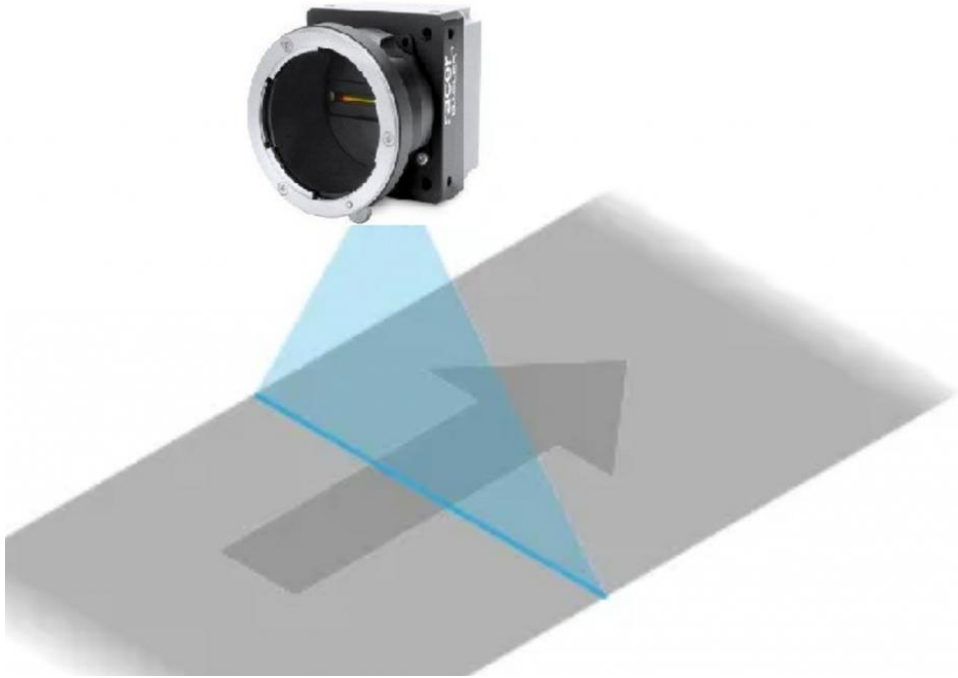
Components



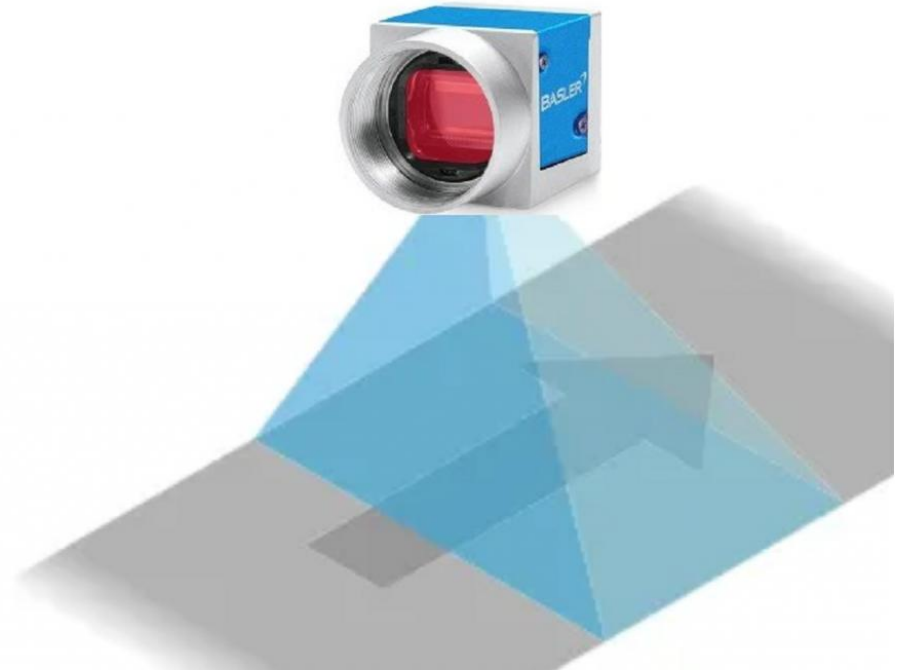
Camera

Camera: Area Scan, Line Scan

Line Scan Camera

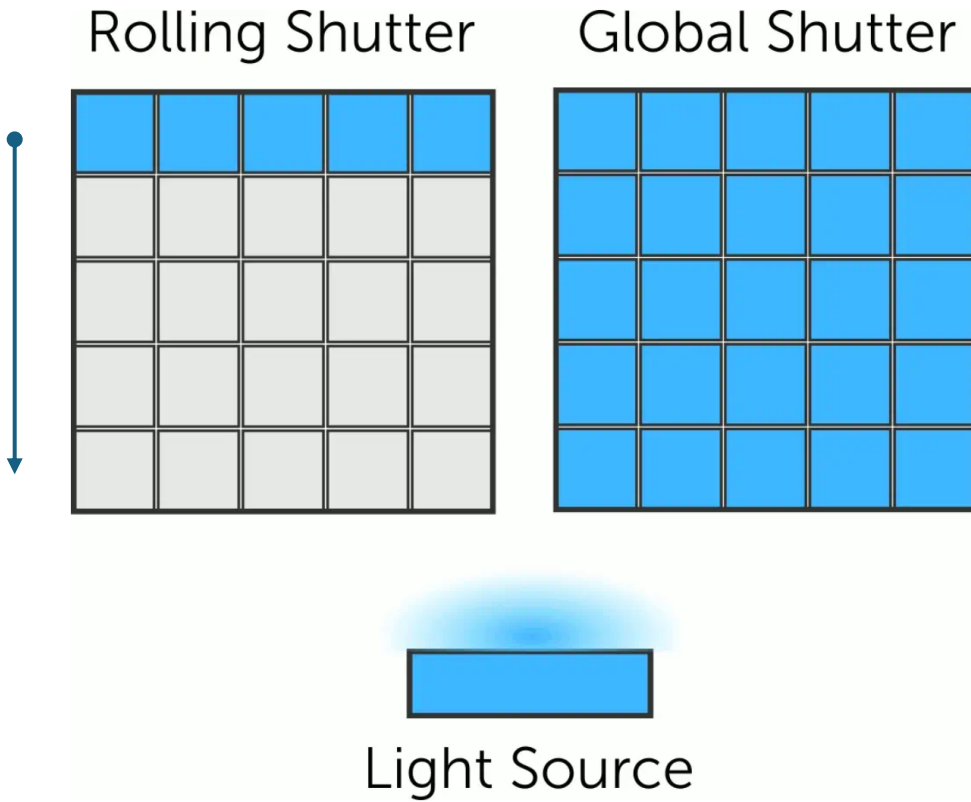


Area Scan Camera



Camera: Rolling Shutter vs. Global Shutter

Concept



Result



Camera: Colour vs. Monochrome



Camera: Interface



CoaXPress Cameras



USB3 Cameras



GigE Cameras



CameraLink Cameras

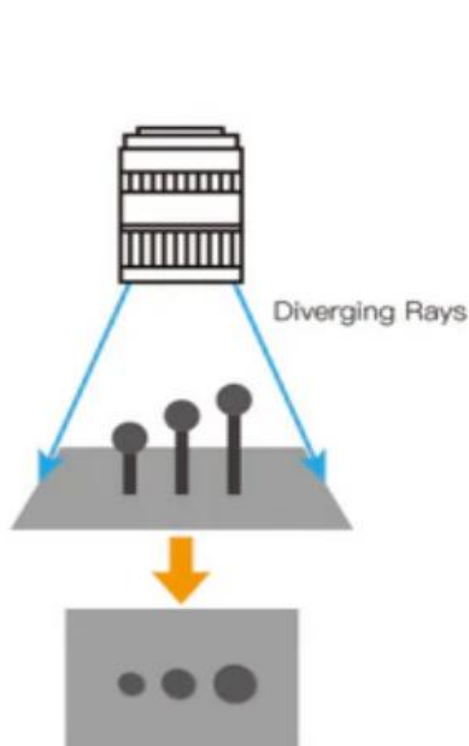


Other Cameras

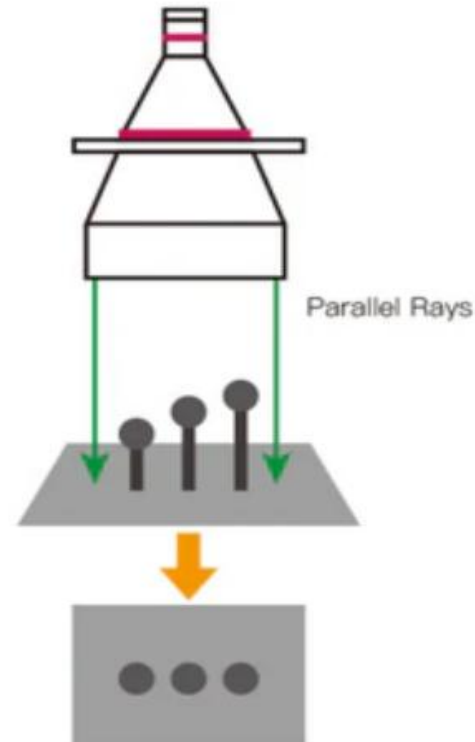
1. How to choose a camera interface? Bandwidth (Image data transferred from camera to the host system.)
2. Cable Length.

Lens

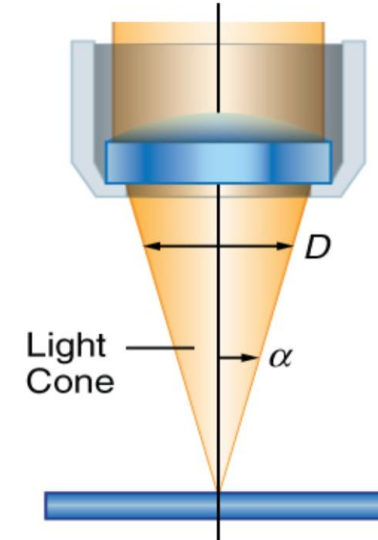
Lens: Fixed Focal Length Lens vs. Telecentric lens vs Objective Lens



Fixed Focal Length Lens



Telecentric Lens

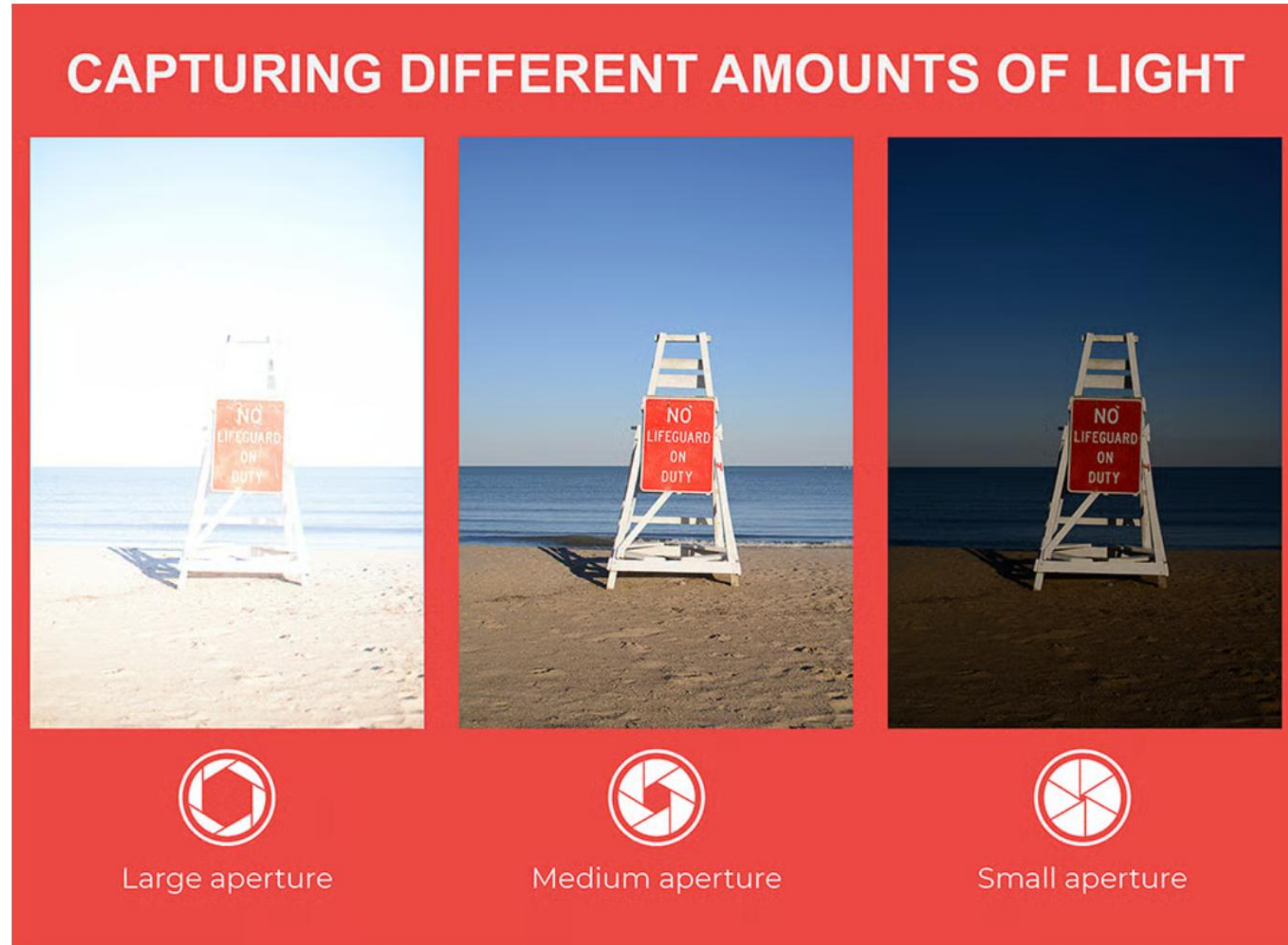


Objective Lens

Reference:

1. [Fixed focal length lens vs. telecentric](#)
2. [Objective lens](#)

Lens: Aperture (Light)



Refence:

1. [Aperture](#)

Lens: Aperture (Depth of Field)



Large aperture



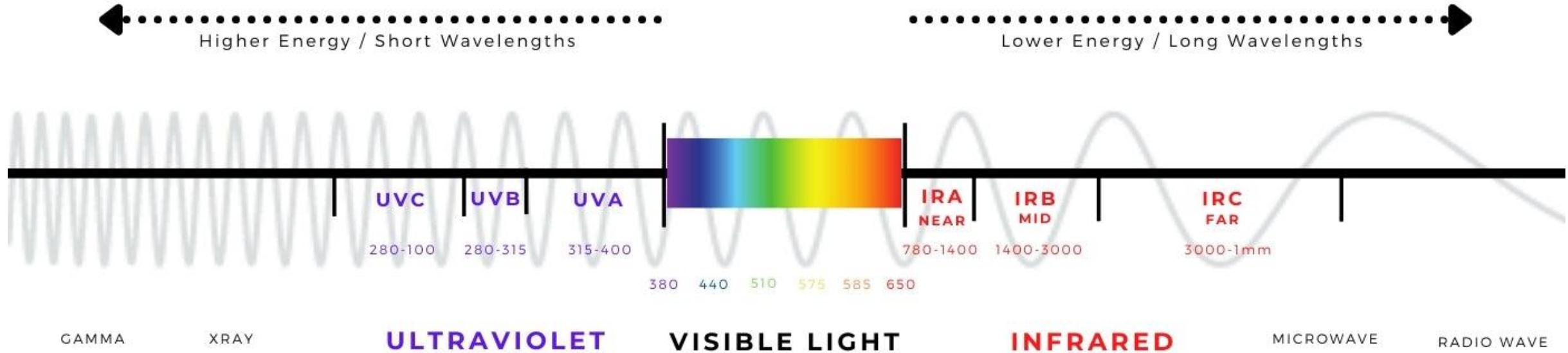
Small aperture

Reference:

1. [Aperture](#)

Light / Lighting

Light: Spectrum of light



Refence:

1. [Spectrum of light](#)

Light: Optical Response

Reflection: Light bounces off the surface.

Absorption: The object converts light into heat.

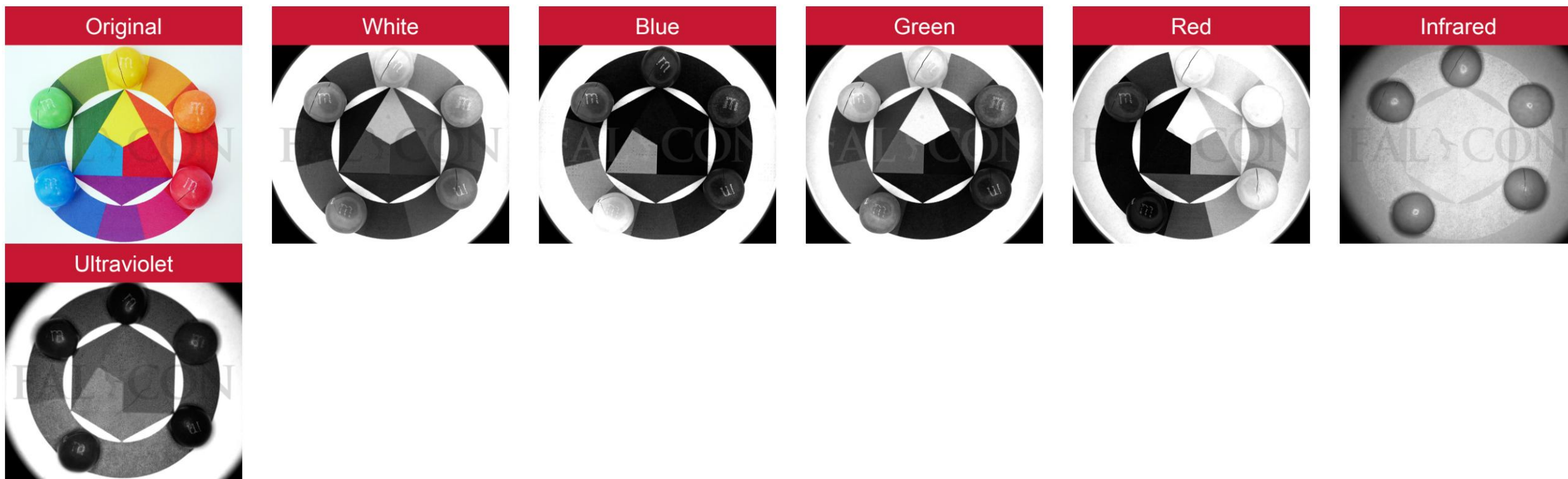
Transmission: Light passes completely through the material.

Refraction: Light bends as it changes medium

Reference:

1. [Spectrum of light](#)

Light: Illumination Colour



Refence:

1. [Illumination Colour](#)

Light: Illumination Angle

Bright Field



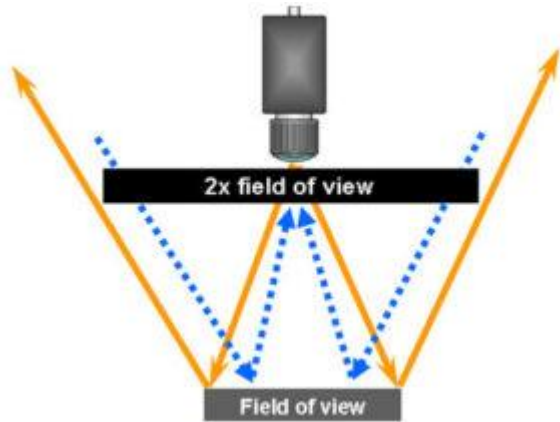
Dark Field



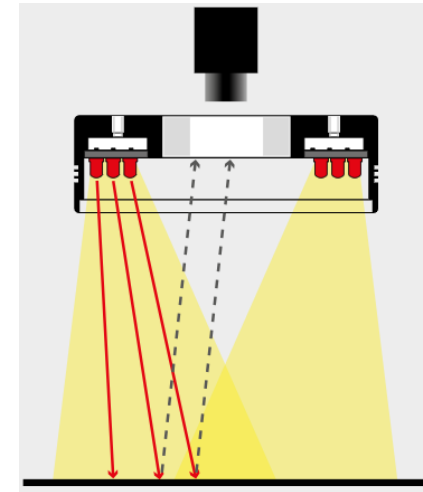
Refence:

1. [Illumination Angle](#)
2. [Bright or Dark Field Lighting in applications](#)

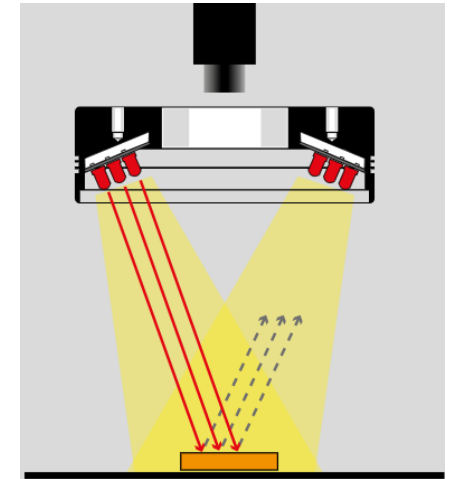
Light: Illumination Angle



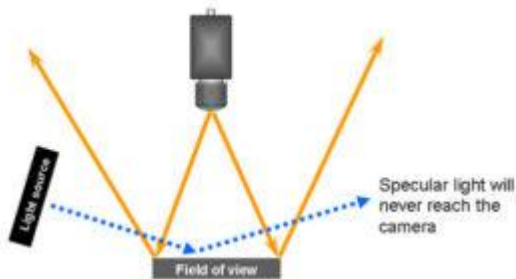
Bright Field Lighting



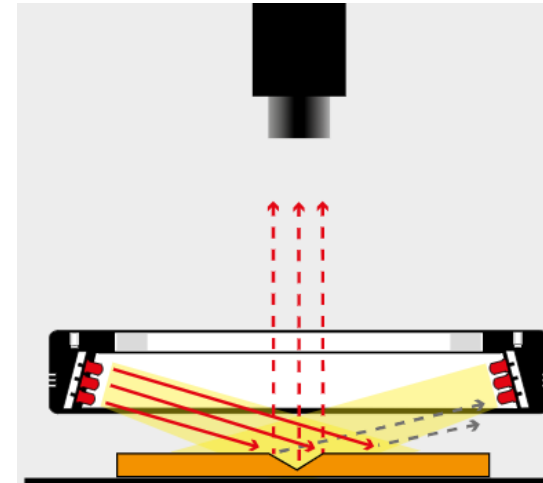
Incident light, direct



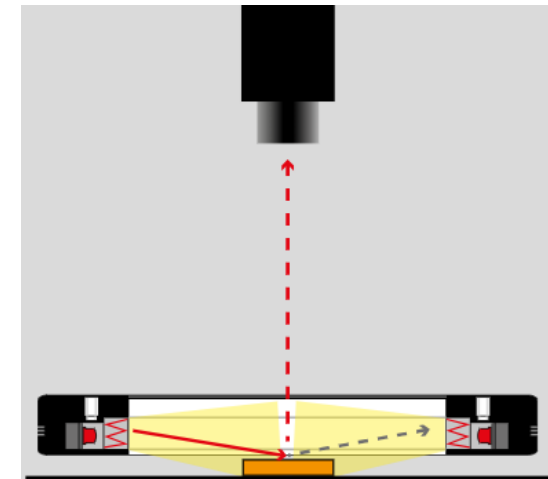
Incident light, angled



Dark Field Lighting



Dark field, tilted



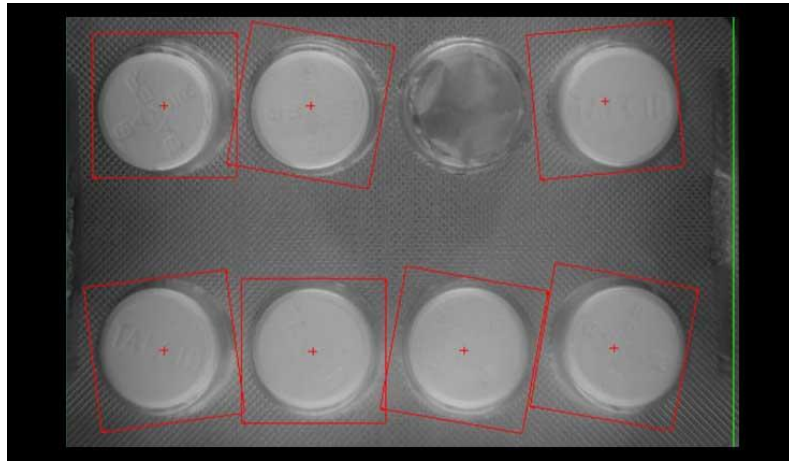
Dark field, horizontal

Reference:

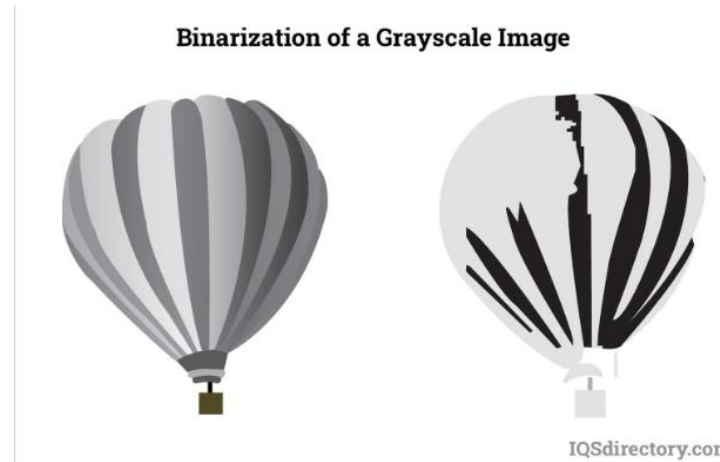
1. [Illumination Angle](#)
2. [Bright or Dark Field Lighting in applications](#)

Image Process Techniques

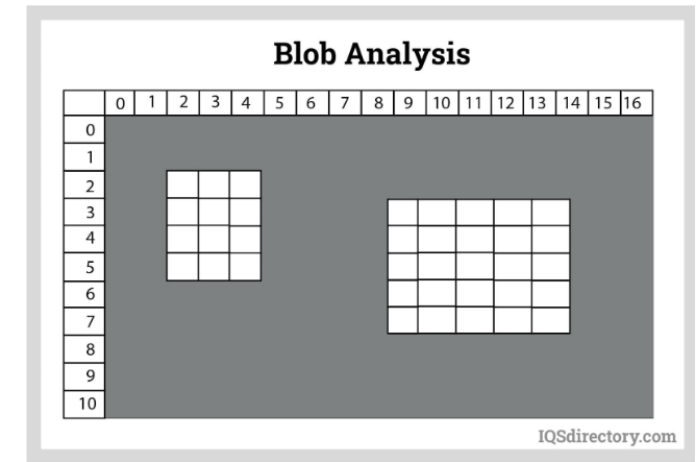
Image Process Techniques



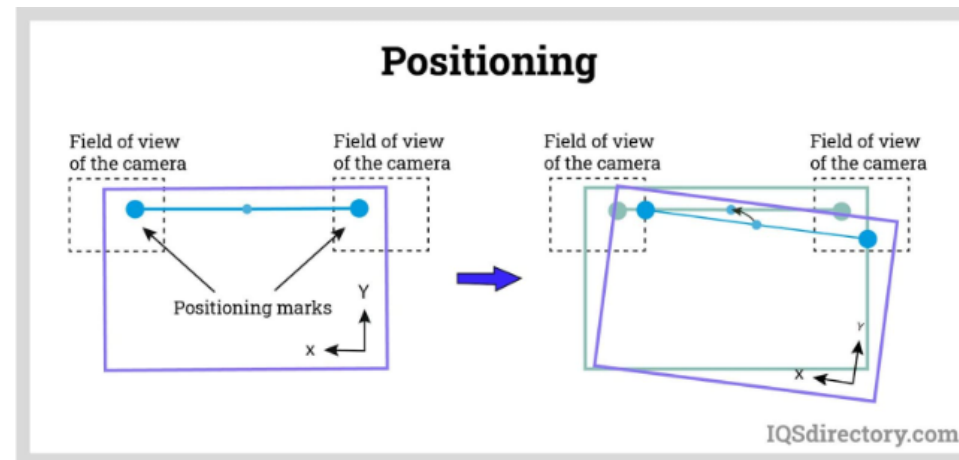
Presence/Absence



Binary Processing



Blob Analysis



Positioning

Reference:

1. [Machine Vision System](#)

Thank you